How Much of an Expected Goals Model Is in the Words?

Recovering shot quality from free-text match commentary

Dheepak Karan Elumalai Santhakumari

Northeastern University, Boston MA, USA
elumalaisanthakuma.d@northeastern.edu

Abstract. Expected goals (xG) models are built on shot coordinates, which are collected by hand or by camera and are largely paywalled. Free-text match commentary is neither: it is published for far more competitions than coordinates are, it needs no key, and it describes each shot in a sentence. We ask how much of a coordinate-based xG model survives if the only input is that sentence.

We parse 87,980 shots from ESPN commentary across six European competitions into eighteen binary and ordinal fields, fit a logistic regression on three Premier League seasons, and evaluate on a held-out fourth (AUC 0.7709). Joining 8,825 shots to StatsBomb's open Premier League 2015/16 data, where both a commentary sentence and a hand-collected coordinate xG exist for the same shot, the text model reaches AUC 0.7826 against StatsBomb's 0.8118: the words recover **90.6%** of the coordinate model's discrimination above chance. Per team-match the two agree at $r = 0.869$, which sits inside the band that commercial providers occupy against each other. Handing a model the whole sentence instead of the extracted fields is worth -0.010 AUC, so the extraction is not the bottleneck; the missing 9.4% is information the sentence never contained.

We then report a failure mode specific to text-derived features. Confirming Robberechts and Davis, a decade-stale model loses only 0.0068 AUC and transfers between leagues at no cost — the *ranking* is stable. The *level* is not: the shipped model over-predicts the held-out season by 11.9%, which decomposes exactly into a 5.7% fall in the league goal rate and a 5.9% shift in phrase mix. Two phrases carry the second half; unambiguous phrases such as "penalty" do not move at all. Retraining cannot fix this, but refitting a single intercept on shots already played reduces the error to 2.9% without touching the ranking. A coordinate does not drift. A phrase does, because a person writes it.

Keywords: Expected goals · Football analytics · Text as features · Concept drift · Calibration

1 Introduction

An expected goals model estimates the probability that a shot becomes a goal. Every widely used version is built on where the shot was taken from: a coordinate pair, a distance, an angle, and increasingly the positions of everyone else in frame [6,3]. Those coordinates are the expensive part. They are produced by human annotators or tracking cameras, and outside a handful of open releases they are sold rather than published.

Free-text commentary is the opposite. For each shot, a sentence like

> *"Attempt saved. Bukayo Saka (Arsenal) right footed shot from the centre of the box is saved in the bottom left corner. Assisted by Martin Ødegaard with a through ball following a fast break."*

is published live, without a key, for competitions that will never have open coordinates. It plainly contains *some* of what an xG model wants. The question this paper answers is: how much?

That framing matters, because the answer is a number rather than a yes or no. If commentary recovers a small fraction of a coordinate model, it is a curiosity. If it recovers most of one, then xG becomes available in every competition where somebody writes sentences about the football — which is most of them.

Contributions

1. **A measurement.** On 8,825 shots for which both a commentary sentence and a StatsBomb coordinate xG exist, a model reading only the sentence recovers **90.6%** of the coordinate model's discrimination above chance (Section 5.2).
2. **A scale for that number.** Agreement with StatsBomb per team-match is $r = 0.869$, comparable to how closely commercial providers agree with each other (Section 5.3).
3. **An upper bound on extraction.** Giving models the full sentence instead of eighteen extracted fields — every 1–4 gram, and a sentence embedding — makes them *worse* by 0.010 AUC. The remaining gap is not a parsing failure (Section 5.4).
4. **A drift result specific to text features.** The ranking is stable over a decade, replicating [9] on a different feature source; the calibration level is not, and we identify the two phrases responsible, show a control phrase that does not move, and give a leak-free in-season correction (Section 6).
5. **A reproducible pipeline.** Collection, parsing, training, evaluation and every figure in this paper rebuild from one script [4].

2 Related Work

Coordinate xG. Expected goals is two decades old and well surveyed. Recent work has focused on explainability [3], player and position correction [10], and demonstrating downstream value [6]. All of it takes location as given.

Data requirements of xG. Robberechts and Davis [9] ask what data an xG model actually needs. They find a basic logistic model converges at roughly 6,000 shots, that training on less recent data costs “only a negligible performance hit”, that league-specific and cross-league models “perform equally”, and that data from different providers should not be mixed. They argue calibration, not ranking, is the objective. This paper is in their debt twice: our staleness and transfer results replicate theirs from a different feature source, and our drift finding is only interesting because they established the coordinate-side baseline.

Language and football. Text has been used to *forecast* events from event sequences [7], and to *explain* an xG model by turning its coefficients into sentences [8]. The latter is this paper’s mirror image: they go from a model to words, we go from words to a model. That both directions are being explored suggests the interface between the two is worth studying; it does not make either one the other.

Prior art on the input. A public hobby project [2] scrapes Premier League website commentary and states directly that “we can infer the shot position from the text commentary”, deriving goal-conversion rates by shot category. To our knowledge this is the first use of commentary text as a shot-quality input, and it predates this work. It reports no fitted model, no held-out evaluation, no comparison against a reference xG model, and no AUC, log loss or Brier score. **The idea is not ours. The measurement is what we contribute.**

Provider agreement. Public comparisons of Opta, Understat, StatsBomb and Wyscout report match-level xG correlations between 0.86 and 0.96 depending on the pair [5]. We use this as the scale against which our own agreement with StatsBomb should be read.

Feature corroboration. American Soccer Analysis publish their xG feature set [1]: distance, goal mouth available, headed, off a cross or through ball, and pattern of play including corners, free kicks, fast breaks and penalties, fitted with logistic regression. With the exception of the two continuous geometric terms, this is almost exactly the field list a commentary sentence yields — independent evidence that the phrase-derived fields are the right ones to want.

3 Data

3.1 Commentary

ESPN publish an Opta-derived English commentary feed through a public endpoint requiring no key. We collected 3,569 matches across six competitions (Table 1), yielding 87,980 parsed shots and 10,109 goals.

Table 1. The commentary corpus. Premier League spans 2015/16 and 2022/23–2025/26; the other five competitions cover 2022/23–2025/26.

Competition	Matches	Shots	Goals	Goal rate
Premier League	1,892	47,431	5,489	11.6%
La Liga	380	9,240	1,024	11.1%
Serie A	380	9,065	922	10.2%
Bundesliga	306	7,853	990	12.6%
Ligue 1	305	7,391	863	11.7%
Primeira Liga	306	7,000	821	11.7%
Total	**3,569**	**87,980**	**10,109**	**11.5%**

3.2 The reference model

StatsBomb release Premier League 2015/16 openly [11], with a true shot location, a freeze frame of every player, and their own xG for each shot. Joining it to ESPN commentary for the same matches is the only way to put words and coordinates on the same shots.

The join is deliberately conservative: a shot pairs only when the match, the team and the minute all agree, and an unmatched shot is dropped rather than guessed at. This yields **8,825 shots across 373 matches**. As a sanity check on the join itself, the two sources agree on whether the shot was a goal for **99.71%** of paired shots (26 disagreements). All evaluation in Section 5.2 uses StatsBomb's goal flag as the label, so their model is not penalised for annotation differences in ours.

4 Method

4.1 The label is in the sentence

Commentary states the outcome before it describes the chance. Every line begins “Goal!”, “Attempt missed”, “Attempt saved” or “Attempt blocked”, and often repeats it in the verb (“is saved in the bottom left corner”, “hits the left post”). That text *is* the label. A model handed the raw sentence scores a perfect AUC and has learned nothing.

Removing it is harder than it looks, and four separate leaks survived earlier attempts:

1. Stripping only the opening clause left penalties at 100% conversion.
2. A blacklist of outcome words missed the verbs; inspection of the fitted coefficients found `goal` weighted at +7.99.
3. After moving to a **whitelist** — keeping only spans matching a fixed list of chance-describing patterns — the phrase “from a free kick” still converted 60 of 60. Free kicks are consequently absent from the feature set.

Table 2. Goal rate by parsed field, 37,929 Premier League shots (2022/23–2025/26). Overall rate 11.8%.

Field	% shots	Goals	Field	% shots	Goals
penalty	1.0	82.7%	from cross	18.6	10.7%
six yard box	5.0	39.6%	assisted	75.9	10.7%
after fast break	2.7	35.5%	header	18.2	10.5%
through ball	3.5	25.7%	difficult angle	2.5	8.1%
after corner	9.8	16.4%	side of box	18.5	5.6%
centre of box	36.2	14.4%	outside the box	30.6	4.1%

4. The `header` field was being set by the phrase "headed *pass*", mislabelling 1,636 shots and inflating their apparent conversion from 10.0% to 13.8%. Fields describing the shot itself are now read only from the text preceding "assisted by".

The lesson generalises: with text features, a leak is a phrasing nobody thought of, so the guard has to be behavioural rather than a word list. We train a model on the stripped text and assert it *cannot* separate goals too well (ceiling AUC 0.85, against 1.00 for the raw text), and we flag any n-gram whose Wilson lower bound on conversion exceeds that of penalties — the best legitimate feature. This is the check that caught leak (3) and cleared a genuine high-conversion phrase ("close range following fast break", 26 of 32) that a fixed 80% threshold had wrongly flagged.

4.2 Features

Eighteen fields are extracted by whitelist regex: six location zones (`six_yard`, `centre_box`, `side_box`, `outside_box`, `long_range`, `difficult_ang`), three body parts, five build-up descriptors (`from_cross`, `from_headed_pass`, `from_through`, `after_corner`, `after_break`), `assisted`, `penalty`, and the minute.

Table 2 shows the resulting spread. A twentyfold range in conversion, recovered from English sentences alone, is the reason the rest of the paper is possible.

4.3 Model and protocol

The shipped model is a standardised logistic regression. It is deliberately the simplest thing that works: with eighteen mostly binary fields there is little for a stronger learner to find, and Section 5.4 confirms this empirically.

Two distinct fits are reported and should not be confused.

– **The shipped model**, trained on Premier League 2022/23–2024/25 (28,735 shots) and evaluated on the held-out 2025/26 season (9,194 shots).
– **The comparison model**, trained on Premier League 2022/23–2025/26 (37,929 shots) and evaluated on 2015/16, which is excluded from its training data. This is the fit used against StatsBomb.

Table 3. Words against coordinates, 8,825 shots, Premier League 2015/16.

Model	Input	AUC	Log loss	Brier
Ours	the commentary sentence	0.7826	0.2711	0.0764
StatsBomb	coordinates, freeze frames, 16 player positions	0.8118	0.2555	0.0715

All training is scoped to the Premier League, following [9] on not mixing sources; the other five competitions are held out entirely as a transfer test (Section 5.5).

5 Results

5.1 Held-out season

On 2025/26, unseen during training, the model reaches **AUC 0.7709**, Brier 0.0840, against a base rate of 11.3%.

5.2 Against a coordinate model

Table 3 is the central result. Both models predict the same 8,825 shots, labelled by StatsBomb.

$$\frac{0.7826 - 0.5}{0.8118 - 0.5} = \mathbf{0.906} \tag{1}$$

The words recover 90.6% of the coordinate model's discrimination above chance. The mean predictions agree to within 0.001 (0.0972 against 0.0982), so whatever the words miss, they do not miss it in a biased direction. Per-shot correlation is 0.735 (Spearman 0.628), mean absolute difference 0.052.

5.3 Is 90.6% close? A scale

A recovery figure means little without something to compare it to. The natural comparison is how closely the commercial providers agree with *each other* [5], and Table 4 places our model on the same axis.

Mean absolute difference is 0.249 xG per team-match (median 0.198, max 1.54); mean xG 1.150 against StatsBomb's 1.161 and 1.192 actual goals.

A model that reads only English sentences agrees with StatsBomb about as closely as a commercial provider does. Two caveats attach. First, the aggregation levels are close but not identical: theirs is per match across five leagues and five seasons, ours is per team-match on one league and one season. Second, a widely repeated "≈1 xG mean absolute difference between providers, maximum 3.88" figure is deliberately *not* used here. Naming a single club alongside 3.88 reads as a season aggregate, which would make that figure far tighter per match than our 0.249 rather than looser — and the source paywalls, so we cannot confirm the unit.

Table 4. Match-level xG agreement. Provider figures from [5] (five leagues, five seasons, ≈4,290 matches); ours computed on 746 team-matches of the join.

Pair	Correlation
Opta × Understat	0.96
Opta × StatsBomb	0.92–0.93
StatsBomb × Understat	0.92–0.93
Wyscout × the others	0.86–0.88
Ours (commentary text) × StatsBomb	**0.869**

Table 5. Reading more of the sentence, held-out 2025/26 season.

Model	Sees	AUC
18 regex fields, logistic (shipped)	18 fields	**0.7709**
18 regex fields, gradient-boosted trees	18 fields	0.7695
every 1–4 gram, TF-IDF	all words	0.7612
sentence embedding (MiniLM)	all words	0.7584
embedding + regex fields	both	0.7589

5.4 Is the extraction the bottleneck? No

The 9.4% gap has two possible homes: the regexes catch only phrasings someone thought of, or the sentence never contained the information. This is decided by handing models the whole sentence and seeing whether they beat the eighteen fields pulled out of it (Table 5).

Every model given the full text does *worse*, including a semantic embedding; adding the embedding to the fields makes those worse too. The rest of the sentence is player names, team names and filler. **Reading more is worth** -0.010 **AUC**, so no better reader — an LLM extractor included — closes the gap. The missing 9.4% is information the sentence never contained: the exact distance inside a zone, the angle, and how many defenders stood in the way.

5.5 Transfer

The claim that xG becomes available wherever commentary exists is only real if the model travels. The Premier League model was pointed at five other competitions cold — nothing retrained, nothing tuned (Table 6).

Mean AUC abroad is 0.7781 against 0.7709 at home: a transfer cost of -0.0072, which is to say a small gain. Calibration holds, with predictions of 0.115–0.125 against observed rates of 10.2–12.6% and the ordering preserved. This replicates [9] on a feature source they did not consider, and it works for a reason specific to this setting: Opta build the sentence the same way in every competition.

Table 6. One model, six competitions. Nothing retrained.

Competition	Shots	Goal rate	AUC	Mean xG
Premier League *(trained on)*	9,194	11.3%	0.7709	0.127
La Liga	9,240	11.1%	0.7730	0.119
Ligue 1	7,391	11.7%	0.7807	0.122
Bundesliga	7,853	12.6%	0.7795	0.125
Serie A	9,065	10.2%	0.7702	0.115
Primeira Liga	7,000	11.7%	0.7871	0.122

Table 7. Phrase drift in Premier League commentary. Two phrases move; two controls do not.

	2015/16		2025/26		
Phrase	% shots	conv.	% shots	conv.	
"six yard box"	3.22	43.5%	5.96	35.8%	**drifts**
"following a fast break"	0.73	49.3%	3.30	28.4%	**drifts**
"centre of the box"	32.84	14.2%	36.44	13.2%	stable
"penalty"	0.94	83.1%	1.01	82.8%	stable

6 The Ranking Holds, the Level Does Not

6.1 The symptom

On the held-out season the model predicts 1.54 goals per team-match against 1.38 actual — an **11.9% over-estimate**. It decomposes exactly:

		Factor
league goal rate, 2022/23 → 2025/26	0.1199 → 0.1134	× 1.057
mean prediction vs. its own training rate	0.1199 → 0.1270	× 1.059
observed over-estimate		**1.119**

The first factor is the league converting slightly less often. The second is stranger: a logistic regression's mean prediction on its training set equals the training base rate by construction, so a mean of 0.1270 on 2025/26 means the *parsed features* of those shots look better than the shots the model was fitted on — while producing fewer goals.

6.2 Two phrases, and a control that does not move

Table 7 localises the second factor. "Following a fast break" became **4.5 times more common at little over half the conversion** in a decade. That is not a change in football. It is a change in how loosely the phrase is written, and the model — which knows nothing except the words — reads a better shot than the one that was taken.

The controls matter as much as the finding. “Penalty” is unambiguous, and it does not move: 0.94% of shots at 83.1% in 2015/16, 1.01% at 82.8% in 2025/26. “Centre of the box” is nearly as stable. **The drift is not a general artefact of the parser or the corpus; it is specific to phrases that require a judgement call from whoever is writing.**

Scored with the shipped model, the sign of the error flips across the decade: −2.7% on 2015/16, +11.9% on 2025/26. The model was fitted in between.

6.3 Retraining does not help

Training window	Over-estimate
all three seasons (shipped)	+11.9%
last two seasons	+11.9%
last season only	+9.1%
recency weighted, half-life one season	+9.9%

The best of these buys 2.8 of the 11.9 points and pays two thirds of the training set for it. The reason is not that the data is old: **next season’s conversion rate is not in this season’s data at any weighting.**

6.4 Waiting does

Once a few hundred shots of the new season have been played, their realised rate is observable, and one number added to the intercept pulls the level back. We walk forward through 2025/26 in blocks of 500 shots, refitting the shift by bisection on *only* the shots played before each block:

	mean xG	actual	over	log loss
uncorrected	0.1267	0.1132	+12.0%	0.29523
recalibrated	0.1165	0.1132	+2.9%	0.29443

8,694 shots evaluated after a 500-shot warm-up — roughly twenty matches — during which the model runs uncorrected because the realised rate is still too noisy to correct towards. An intercept shift is monotone, so AUC is unchanged by construction, and our test suite asserts it: the guard is against improving calibration by reaching for the coefficients, which would trade away the ranking to buy a better mean.

6.5 Why this is not a contradiction of prior work

Robberechts and Davis [9] put calibration at the centre of xG evaluation and found staleness negligible. We reproduce their ranking result exactly — a decade-stale model costs 0.0068 AUC — and observe the level moving 10% over the same span. Both are true, and together they say something neither says alone:

A coordinate does not drift. A phrase does, because a person writes it. Event-data xG is stable in precisely the sense — calibration — where a text-derived model is not, and the instability is confined to phrases requiring human judgement.

This is, to our knowledge, the first characterisation of concept drift in text-derived sports features, and it has a practical consequence: any system built on commentary phrasing needs an in-season level correction, and does not need retraining.

7 Limitations

We state these in full rather than in passing.

- **One reference model, one season.** The headline compares against StatsBomb alone, on Premier League 2015/16. A second provider, or a second season, would strengthen it considerably. StatsBomb's open releases are the constraint.
- **Free kicks are absent.** The phrase "from a free kick" leaked (60 of 60 conversions), and rather than repair it we removed the feature. A model that handled free kicks properly would likely do better.
- **Matches with no commentary are dropped silently.** A small number of matches arrive with an empty feed and contribute nothing. The rate is low but it is not zero and it is not currently guarded.
- **Latency is unmeasured.** We claim commentary arrives fast enough for live use and have no measurement to support it. A scheduled matchday watcher exists but the cloud scheduler that triggers it fired 2 of 12 slots on the day tested, and missed the kickoff.
- **The novelty claim rests on an imperfect search.** Our literature search of arXiv, OpenAlex and Crossref returned 5 of 6 recall probes and not always the same five. A follow-up blog and grey-literature search — which found [2] — returned 3 of 4 probes by name, and `reddit.com` refuses our crawler entirely, which is exactly where an unpublished version of this would be discussed. Both searches are recorded query-by-query in the repository. We have not searched paywalled indexes, thesis repositories, or non-English sources.
- **The provider-agreement comparison is close but not exact**, as noted in Section 5.3.

8 Conclusion

A sentence of English recovers 90.6% of a tracking-data expected goals model's discrimination above chance, and agrees with it per team-match about as closely as one commercial provider agrees with another. The gap is not a parsing failure — richer readers of the same sentence do worse — but information the sentence never contained. The model transfers across six competitions at no cost, because the sentence is built the same way in each.

The interesting failure is not in the ranking but in the level. Text features drift where coordinate features do not, because a phrase is written by a person whose habits change, and we localise that drift to phrases requiring judgement while unambiguous phrases stay fixed. One intercept, refitted in season on shots already played, corrects it.

For competitions that will never have open coordinates — which is most of them — this is the difference between having an xG model and not having one.

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Reproducibility

Every number in this paper is produced by `rebuild.sh` in [4], which runs the pipeline in dependency order from raw collection to final figures. The repository additionally contains a test suite of 50 tests, including behavioural leak guards, a walk-forward check that the recalibration uses no future shots, an assertion that the shift leaves AUC unchanged, and a dependency graph that fails if any artefact predates its inputs. The literature and blog searches are recorded query-by-query with their recall probes.

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